

VACCUUM CLEANER

CODE :

```
import time
status = {'A': 'd', 'B': 'd'}
agent_location = 'A'
print("=== Starting State ===")
print(f"[{agent_location}, {status['A']}, {status['B']}]")
print("-" * 30)
step = 1
while not (status['A'] == 'c' and status['B'] == 'c'):
    print(f"\nStep {step}:")
    current_status = status[agent_location]
    if current_status == 'd':
        action = 'pick'
        status[agent_location] = 'c'
    elif agent_location == 'A':
        action = 'MoveRight'
        agent_location = 'B'
    elif agent_location == 'B':
        action = 'MoveLeft'
        agent_location = 'A'

    print(f"  Action: {action}")
    print(f"  State:  [{agent_location}, {status['A']}, {status['B']}]")

    step += 1
    time.sleep(0.5)
print("\n" + "-" * 30)
print("=== Final State: All Clean! ===")
print(f"[{agent_location}, {status['A']}, {status['B']}]")
```

OUTPUT:

```
**  === Starting State ===  
    [A, d, d]  
    -----  
  
    Step 1:  
      Action: pick  
      State:  [A, c, d]  
  
    Step 2:  
      Action: MoveRight  
      State:  [B, c, d]  
  
    Step 3:  
      Action: pick  
      State:  [B, c, c]  
  
    -----  
    === Final State: All Clean! ===  
    [B, c, c]
```